

Banking campaigns comparator

From public product pages to evidence-backed comparisons

A reusable research workflow

Capture evidence. Structure observations.
Review automatic suggestions. Compare like with like.

Business question: what can ING learn from how other banks present similar products?

Built for human review

React + FastAPI + PostgreSQL
Playwright page capture
Rule-based assistance
No AI model required

The brief asks for evidence, not a bank ranking

A focused, reproducible POC for business and data stakeholders.

Compare consistently

- Same product category
- Clear feature definitions
- Explicit sample and language

Explain differences

- Text and messaging
- Visual design and structure
- Calls to action and value

Make it reusable

- Preserve original evidence
- Save structured labels
- Document limits and assumptions

The research workflow is implemented

Business conclusions still depend on a reviewed, comparable sample.

Implemented

- Page capture and history
- Structured feature framework
- Automatic draft suggestions
- Manual review and completion
- Category-based comparison

Partially addressed

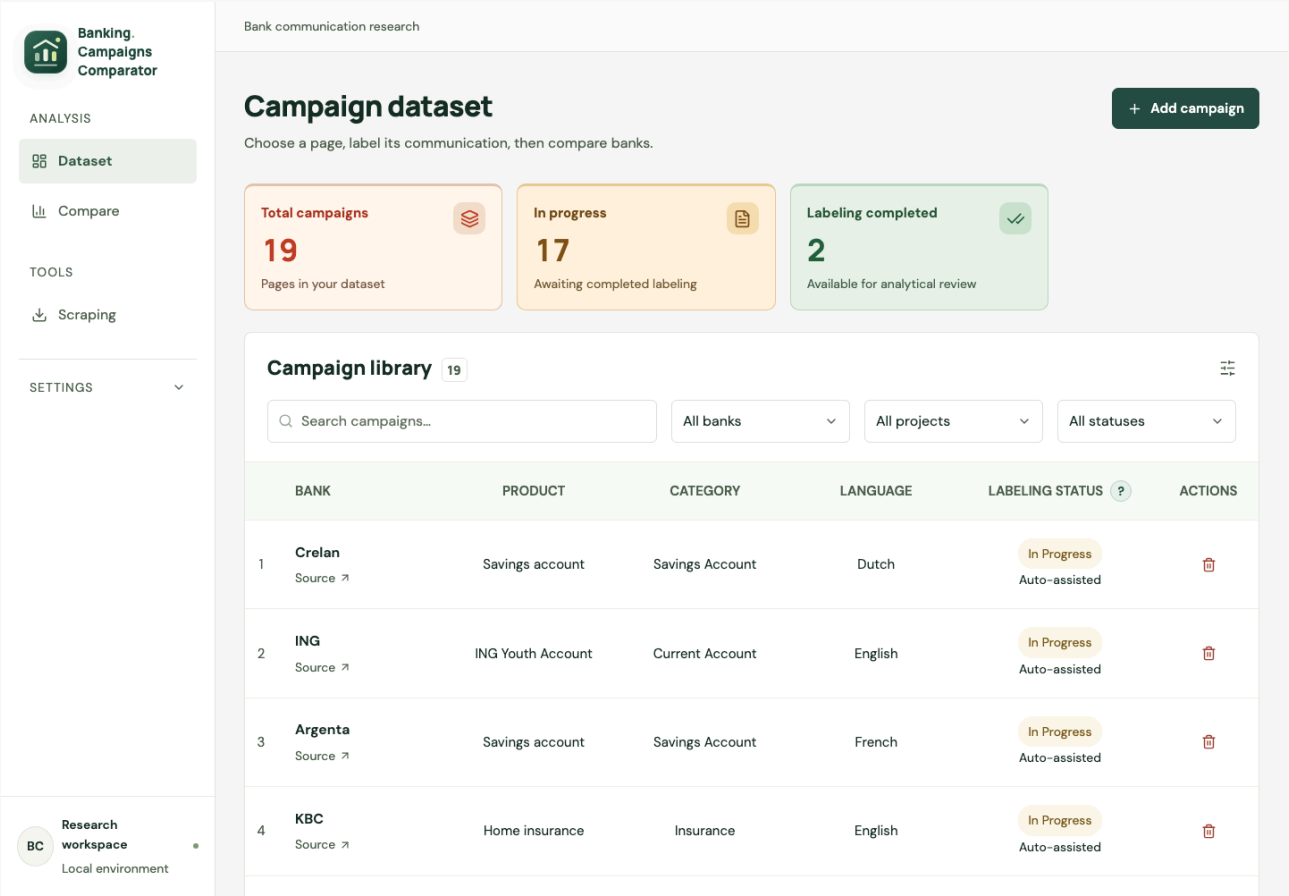
- Limited registered auto cases
- Visual dimensions: manual
- Keyword method calibration
- Sitemap discovery tooling
- Research sample completion

Still to demonstrate

- ING market positioning
- Traditional vs neobank patterns
- Evidence-backed improvements
- Cross-reviewer agreement
- Business impact validation

One workspace for the research team

Actual local app snapshot: 19 campaigns, 2 completed; 17 awaiting completion.



Four working areas

- Scraping sources
- Dataset and filters
- Product review / labeling
- Comparison workspace

```

graph TD
    subgraph INTERFACE [INTERFACE]
        React[React]
        S[Scraping Sources  
Add URL / choose action]
    end

    subgraph PROCESSING [PROCESSING  
FastAPI / Python]
        SA[Source API  
Import / add sources  
Select capture mode]
        SR[Support registry  
Exact bank / category  
product / language]
        PC[Playwright capture  
Render public page  
Save original evidence]
        FE[Feature extraction  
Text / message / CTA  
Registered auto cases]
        CA[Comparison API  
Same product category  
Saved feature values]
    end

    subgraph DATA [DATA  
PostgreSQL + files]
        Sources[Sources  
user_sources / imports  
Configured JSON input]
        CH[Capture history  
PageCapture in database  
Screenshot + dom.json files]
        FR[Feature records  
CampaignFeature  
Notes and review status]
        RC[Research context  
Campaign / bank / category  
Stored in PostgreSQL]
    end

    S -.-> SA
    SA --> Sources
    SA -.-> SR
    SR -.-> PC
    PC -- "Auto-label path only" --> FE
    PC -- "Every successful capture" --> CH
    FE -- "Evidence" --> DL[Dataset & Labeling  
Review evidence and labels]
    FE -- "First auto draft" --> FR
    DL -- "Review & save" --> FR
    FR --> CA
    RC --> CA
    CA --> CW[Compare workspace  
Charts, tables, observations]

    subgraph ACTION_RULES [ACTION RULES]
        AR[First + supported: Scrape & Auto-label | First + unsupported: Capture page | Already captured: Capture only]
    end

```

INTERFACE
React

Scraping Sources
Add URL / choose action

PROCESSING
FastAPI / Python

Source API
Import / add sources
Select capture mode

Support registry
Exact bank / category
product / language

Playwright capture
Render public page
Save original evidence

Feature extraction
Text / message / CTA
Registered auto cases

Comparison API
Same product category
Saved feature values

DATA
PostgreSQL + files

Sources
user_sources / imports
Configured JSON input

Capture history
PageCapture in database
Screenshot + dom.json files

Feature records
CampaignFeature
Notes and review status

Research context
Campaign / bank / category
Stored in PostgreSQL

Compare workspace
Charts, tables, observations

Dataset & Labeling
Review evidence and labels

ACTION RULES
First + supported: Scrape & Auto-label | First + unsupported: Capture page | Already captured: Capture only

Recapture adds evidence and preserves saved labels.

Three actions, chosen by source state

Exact automatic support: bank + category + product + language.

Product URL >> Source in database >> Successful capture already exists?

No + auto supported

Scrape & Auto-label
Evidence + automatic draft

V

No + no auto support

Capture page
Evidence; labels stay manual

V

Yes: already captured

Capture only
New evidence; labels preserved

V

PageCapture history >> Human review >> Saved features >> Compare

Capture the page before analysing it

Screenshot, DOM and extracted content stay available for review.

Bank communication research

ANALYSIS

Dataset

Compare

TOOLS

Scraping

SETTINGS

BC

Research workspace

Local environment

Bank communication research

Scraping Sources

Add source

Add product sources, capture evidence, and review labels.

Bank

Product category

Product

Language

Status

All banks

All categories

All products

All languages

All statuses

Select all

Clear

Scrape selected (0)

BANK / PRODUCT	CATEGORY	LANGUAGE	AUTOMATION	STATUS	URL	ACTIONS
☐ Belfius teenager acc	Current Account	English	Generic capture only	Imported	www.belfius.be View campaign → View captures	Capture only
☐ Belfius normal acc	Current Account	English	Generic capture only	Imported	www.belfius.be View campaign → View captures	Capture only
☐ Belfius credit card	Credit Card	French	Generic capture only	Ready	www.belfius.be	Capture page
☐ ING ING Youth Account	Current Account	English	Specialized scraper available	Imported	www.ing.be View campaign → View captures	Capture only

Evidence saved

Full-page screenshot
DOM / shadow-root data
Text and headings
Paragraphs, lists, tables
Timestamp and warnings

68 fields across 8 analytical sections

Plus campaign observations and labeling notes for research context.

01 / 4 fields

Identification & Metadata

02 / 9 fields

Text & Content

03 / 9 fields

Messaging & Tone

04 / 12 fields

Images & Visuals

05 / 7 fields

Colour & Design

06 / 10 fields

Layout & Page Structure

07 / 6 fields

Call to Action

08 / 11 fields

Product & Value Proposition

Measurements and heuristics produce a draft

Automation supports reviewers; it does not determine effectiveness.

Text measurements

- Word and heading counts
- Paragraph and bullet counts
- Headline length
- Case-specific text heuristics

Message and tone

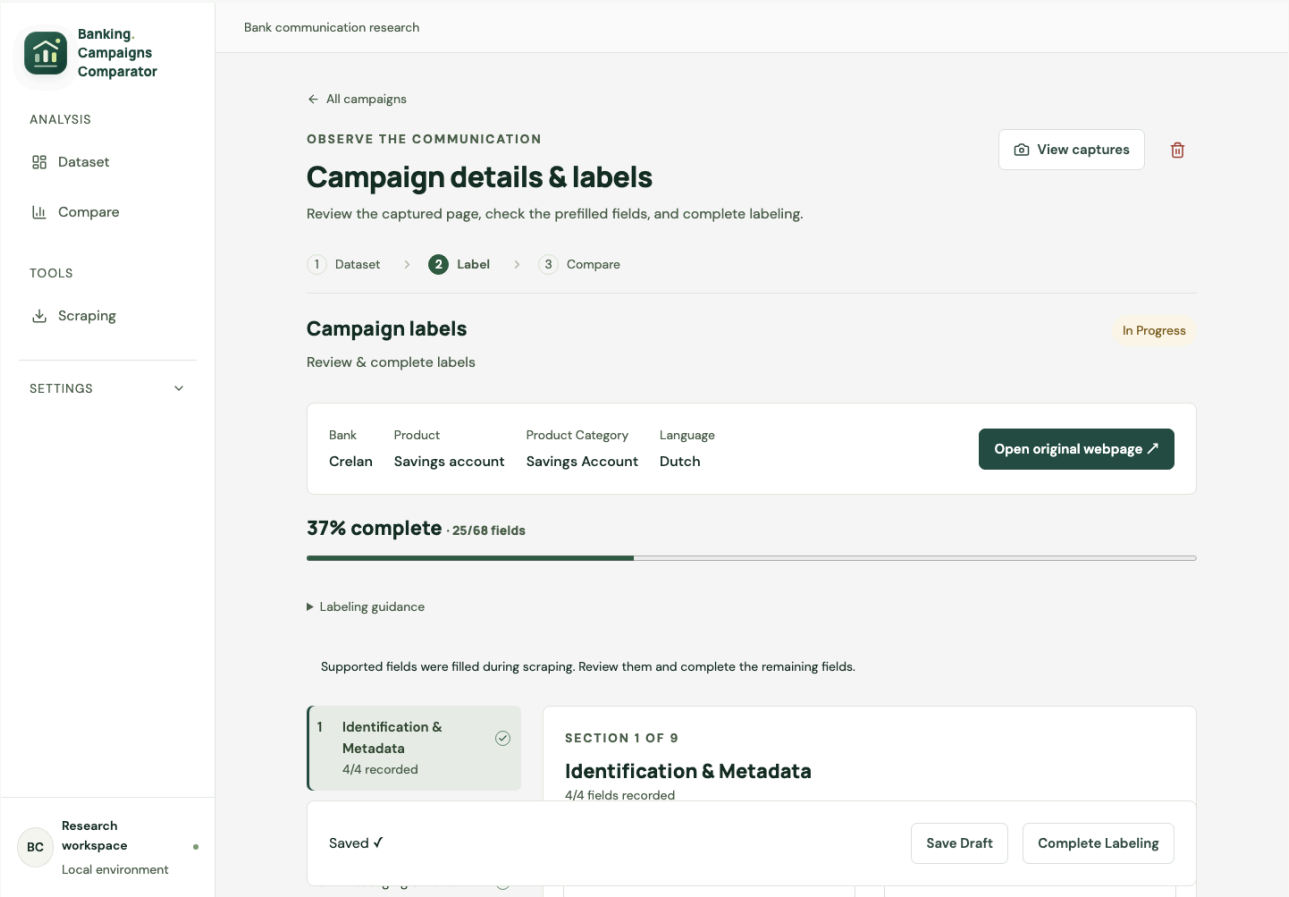
- Feature vs benefit focus
- Emotional vs rational
- Formality and friendliness
- Keyword-based scale scores

CTA analysis

- Count and primary text
- Above-fold placement
- Repeated primary CTA
- Prominence (1?5) and type

Keep the analyst in control

Not Started >> In Progress >> Completed

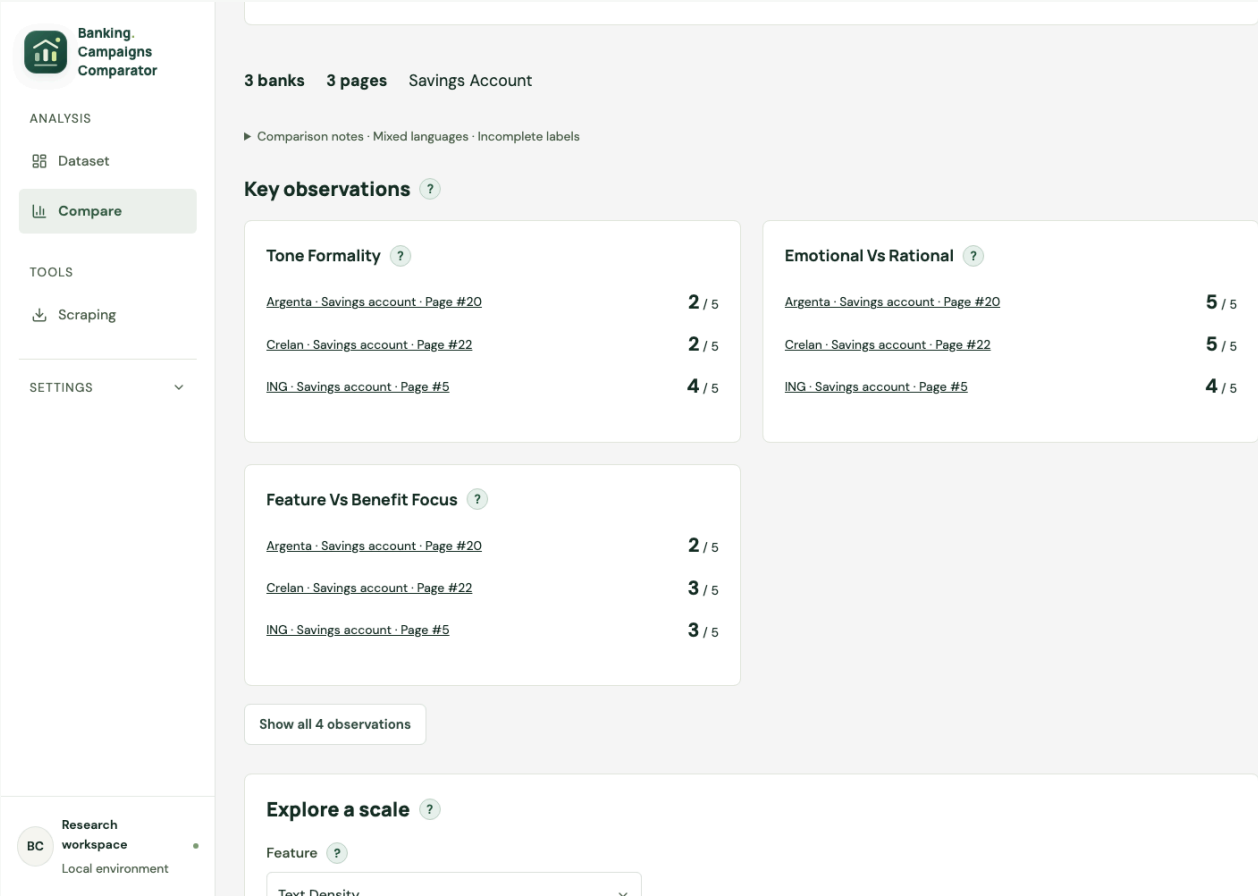


Reviewer decisions

- Check captured evidence
- Correct suggested values
- Label visual dimensions
- Record notes and context
- Complete the review

Compare recorded page characteristics

One category at a time; each page remains a separate observation.



What the views show

- Selected-page feature tables
- Descriptive 1?5 charts
- Ranges and coverage
- Missing-data warnings
- Links back to product review

Useful POC; validation is still essential

Technical checks and research validity answer different questions.

Engineering evidence

192 tests passed in the last run
CTA browser fixture checks
Saved feature / recapture tests
Generic and supported paths
No production-readiness claim

Methodology risks

Vocabulary and language bias
Page length and text scope
Cookie / dynamic content
Manual reviewer differences
Small, incomplete sample

Collection boundaries

Public-page collection only
URL validation in generic flow
Sitemap discovery available
Review website permissions
No claim of universal access

What ING can use ? and what comes next

A repeatable comparison process is delivered; conclusions need reviewed evidence.

Value delivered

- Less fragmented evidence
- A common feature vocabulary
- Traceable review decisions
- Repeatable comparison views

Before final conclusions

- Match category and language
- Include relevant competitors
- Complete and double-review
- Validate keyword scores

Next business output

- Explain 2?3 real differences
- Show evidence for each claim
- Position ING within the sample
- Propose testable improvements

Show the full loop in three minutes

Source >> Capture evidence >> Review labels >> Compare

01 Source

Show support and action state

02 Evidence

Open screenshot and DOM history

03 Review

Explain a CTA value and its provenance

04 Compare

Read a difference and its coverage

Discussion: which comparable sample should we validate next?

Evidence behind this presentation

Requirement sources and implementation references for Q&A.

Stakeholder requirements

20260914_ProjectPresentation-compressed.pdf

Pages 19-22: comparisons, scope and POC

use_case_description_beCode_campaign_comparator.docx

Objectives, features and expected outputs

Implementation evidence

scraping/README.md and scraping modules

services/collection.py and services/compare.py

models/features.py and frontend/api/features.ts

Scraping, Label and Compare screens

Backend tests and frontend/styles.css